Is the Platform Part of the Measurement?
Testing Cross-Platform Equivalence of Machine-Labelled Policy Sentiment

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Abstract

Comparing machine-labelled sentiment across social media platforms is now routine: differences in classifier scores between platforms are read as differences in public opinion. These comparisons rest on an assumption that the classifier measures the same construct, on the same scale, on every platform. To our knowledge, no study has tested that assumption for machine-labelled constructs by asking whether the platform is part of the measurement. We adapt the psychometric equivalence toolkit (generalizability theory, measurement invariance and differential item functioning, and score linking) to machine-labelled policy sentiment, treating the platform as a survey mode. We apply it to [TODO: approximately 279,000 posts; confirm totals and the collection window after recount: per-platform sums reach roughly 395,000 and the results figures classify 255,446 items] about UK economic policy from Twitter/X, Reddit, YouTube, and Quora [TODO: collection window, pending recount], labelled by a fine-tuned BERT classifier and validated against a probability-sampled human-annotated criterion [TODO: to be drawn; the dissertation-stage criterion was 200 balanced posts, kappa approximately 0.78]. A same-event design controls topic and time, pre-identified fiscal events centred on the September 2022 mini-budget provide the linking anchors, and replication with a non-lexicon transformer separates platform mode effects from lexicon artefacts. [TODO: One-sentence headline invariance result.] [TODO: One-sentence naive pooled index versus measurement-adjusted index result.] The protocol lets a researcher test whether pooling platforms is defensible in a given corpus, or whether platform scores must be linked before any average is taken.

1 Introduction

Computational social science now compares machine-labelled text across platforms as a matter of course. Overgaard et al. [2024] apply a BERT classifier to Facebook and Twitter posts and read the difference in scores as a difference in behaviour. Wei et al. [2026] run the same design across Douyin and TikTok. Studies in this mould publish a cross-platform comparison, or a pooled multi-platform index, on the assumption that the classifier behaves identically wherever it is pointed. The assumption is rarely stated, still more rarely examined, and it is not an obviously safe one. Nogara et al. [2025] report that Perspective API scores German comments as more toxic than equivalent English ones: the instrument behaves differently in different languages. In Atreja et al. [2025], LLM annotations shift under changes of prompt wording alone. And Greene et al. [2025] find that the same political actors are read differently on different platforms. Classifiers are instruments, and their behaviour depends on the conditions in which they are used.

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When the labels are machine-produced, nobody asks with a formal test whether the platform is part of the measurement. When a classifier scores Reddit discussion of a budget more negative than tweets about the same budget, the difference is read as a fact about opinion, when it could equally be a property of the instrument. Character limits, anonymity, threading, moderation, and audience shape how the same attitude is expressed, and therefore how it is scored. Sen et al. [2021] give this hazard a name, platform-affordance error, in their total error framework for digital traces, but they provide no formal test for it. Recent correction methods repair label error against a human gold standard [Egami et al., 2023] and probe the fragility of LLM annotation pipelines [Baumann et al., 2025, Halterman and Keith, 2026], yet none of them models the platform as a source of measurement error.

Survey methodology met a structurally identical problem decades ago and built machinery for it. The same respondent answers differently by telephone, on paper, and face to face; the field calls this a mode effect and does not treat cross-mode differences as substantive until measurement equivalence has been established [Tourangeau et al., 2000, Klausch et al., 2013, Hox et al., 2015], within a total-error accounting of where distortions enter [Groves and Lyberg, 2010]. Our claim is that a platform is a mode. The analogy has limits: absent random assignment of users to platforms or within-user cross-posting, the design identifies platform-level non-equivalence, a bundle of selection and measurement that Section 8 unpacks, with mode as the leading interpretation rather than an established fact. Affordances stand to posts as interview settings stand to answers, and the psychometric toolkit built for modes and groups (generalizability theory [Cronbach et al., 1972, Brennan, 2001], measurement invariance testing [Meredith, 1993, Putnick and Bornstein, 2016], and score linking [Kolen and Brennan, 2014]) transfers to machine-labelled constructs with the platforms as the groups. So far as we can find, no study has carried out that transfer: the toolkit has reached survey self-reports [Boyd et al., 2024], annotator facets [Kennedy et al., 2020, Sachdeva et al., 2022], and single-platform text scales [Pokropek, 2026], but the platform itself has stayed an unmodelled stratum.

The question is also practical. Since the closure of the major research APIs, few teams can study one platform at depth; instead they assemble patchworks of whatever platforms remain accessible, with heterogeneous coverage and access terms [Davidson et al., 2023, Goanta et al., 2026]. Every such patchwork implicitly pools or compares scores across platforms. If a unit of classifier score does not mean the same thing on Reddit as on X, the pooled trend lines and cross-platform contrasts now filling the literature carry an error term of unknown size.

This paper makes four contributions.

1. We formalise the platform-as-mode question for machine-labelled constructs and give what we believe is the first application of the psychometric equivalence toolkit to machine-classified policy sentiment with platform as the grouping facet.

2. We specify a six-step protocol: same-event divergence testing; a fixed-facet generalizability study with a human-validated criterion; categorical measurement invariance and differential item functioning under effect-size criteria; multilevel decomposition of platform effects, with platform contrasts reported both with and without reaction latency; anchor-based score linking on pre-identified fiscal events, centred on the September 2022 mini-budget; and a simulation contrasting a naive pooled index with a measurement-adjusted one. All tests are replicated with a non-lexicon transformer, so only convergent non-invariance is read as a mode effect [Czarnek and Stillwell, 2022], and invariance results are treated as diagnostic evidence rather than grounds to pool [Robitzsch and Lüdtke, 2023].

3. We apply the protocol to [TODO: approximately 279,000 posts; confirm per-platform totals

and the collection window after recount: Ch5 platform sums reach roughly 395,000 (pp. 21-24) against the 279,000 headline (p. 25), and the Ch7 figures classify 255,446 items (p. 36)] about UK economic policy on Twitter/X, Reddit, YouTube, and Quora, labelled by a fine-tuned BERT classifier and validated against a probability-sampled human-annotated criterion [TODO: to be drawn; the dissertation-stage criterion was 200 balanced posts, kappa approximately 0.78].

4. [AWAITING Empirical findings: whether, where, and by how much cross-platform equivalence fails, and what the naive-versus-adjusted contrast implies for pooled indices.]

The paper proceeds as follows. Section 2 locates the problem between computational text measurement and survey mode-effect research. Section 3 describes the corpus, the anchor events, and the validation design. Sections 4 to 7 then execute the six steps in order: establishing divergence, the generalizability, invariance, and multilevel analyses, linking, and the naive-versus-adjusted comparison, with results reported in place as each step completes. Section 8 states the limitations, Section 9 sets out the ethics position, and Section 10 concludes.

2 Related Work

Mode effects in survey measurement. Survey methodology settled long ago that the instrument is part of the measurement. Answers to the same question differ by mode of administration because each mode changes the cognitive and social conditions of answering [Tourangeau et al., 2000]. Klausch et al. [2013] showed that telephone, face-to-face, and web modes yield non-equivalent measurement of identical attitude scales, and mixed-mode designs now model mode explicitly instead of ignoring it as a nuisance [Hox et al., 2015]. The Total Survey Error framework gives these effects a named place in the error budget [Groves and Lyberg, 2010]. Sen et al. [2021] transported this thinking to digital traces and explicitly identified platform-affordance error as the trace analogue of mode error, but the framework names the error without supplying a test for it. Our premise is simply that platform should receive the same formal treatment for machine-labelled trace data that mode receives for surveys.

Measurement for machine-labelled text. A second literature treats classifiers as measurement instruments. Jacobs and Wallach [2021] argue that computational pipelines operationalise unobservable constructs through measurement models whose validity must be tested, and Halterman and Keith [2026] extend this framing to large language models. Empirical work in this vein audits the instrument itself. Pellert et al. [2022] validate sentiment macroscopes against self-report. Atreja et al. [2025] show that model-generated labels shift with prompt design in unpredictable ways, and Baumann et al. [2025] show that defensible configuration choices alone can flip downstream conclusions. Egami et al. [2023] correct downstream estimates for classifier error with design-based supervised learning, but the correction targets label error against a gold standard and leaves platform-specific distortion untouched. In all of this work the classifier is scrutinised; the platform it reads from is not.

Cross-platform studies and assumed equivalence. Multi-platform designs have become standard, and the post-API turn has made heterogeneous data regimes the norm [Davidson et al., 2023, Goanta et al., 2026]. Overgaard et al. [2024] apply a single BERT classifier across Facebook and Twitter and compare the resulting scores directly; Wei et al. [2026] compare Douyin and TikTok in the same way. There is growing evidence that this is risky. Greene et al. [2025] find that the same actors are read differently across platforms, and Nogara et al. [2025] show that Perspective API scores German-language text as more toxic, a measurement artefact that masquerades as a substantive difference. The field's classic cautions concern populations and samples: who is on the platform [Tufekci, 2014], how the stream is sampled [Morstatter et al., 2013], whether models generalise beyond their training population [Cohen and Ruths, 2013], and whether text sentiment tracks opinion at all [O'Connor et al., 2010]. The complementary question, whether the same score means the same thing on Reddit as on YouTube, is routinely assumed and almost never tested.

The psychometric toolkit and nearest misses. The tools for this question already exist in psychometrics. Generalizability theory partitions score variance among measurement facets [Cronbach et al., 1972, Brennan, 2001]; measurement invariance testing asks whether indicators relate to a construct identically across groups [Meredith, 1993, Putnick and Bornstein, 2016]; linking places scores from different instruments on a common scale [Kolen and Brennan, 2014]; and invariance results are read as diagnostic evidence rather than permission to compare [Robitzsch and Lüdtke, 2023]. Several recent studies come close to combining this toolkit with platform data without quite doing so. Boyd et al. [2024] do test cross-platform invariance, though of survey self-reports rather than machine labels. Faceted Rasch measurement has reached annotation through Kennedy et al. [2020] and Sachdeva et al. [2022], with annotators as the modelled facet and platform left out of the model. In Pellert et al. [2022] validation is by correlation alone, with no invariance structure. The multitrait-multimethod models of Cernat et al. [2025] compare survey with digital-trace measures rather than platform with platform. Han and Anderson [2026] document platform-dependent selection and measurement bias, but in online review ratings rather than machine-labelled text. Pokropek [2026] fit factor models to text indicators while grouping by time within a single platform. And generalizability designs have reached LLM annotation, with prompt and model choice as variance facets, without yet reaching platform [Camuffo et al., 2026].

To our knowledge, no study has applied the psychometric equivalence toolkit of generalizability, invariance, and linking to machine-classified policy sentiment with platform as the grouping facet, and this paper supplies that test.

3 Data and Validation Design

Corpus. The corpus contains [TODO: approximately 279,000 (dissertation p. 25); confirm after recount] public posts about UK economic policy, collected from Twitter/X (licensed third-party archive, January 2018 to July 2025; dissertation p. 21), Reddit (official developer API, no stated date restriction; p. 22), YouTube (Data API v3, top-level comments on major UK news and government channels; pp. 22-23), and Quora (the same licensed analytics partner as Twitter; p. 23). [TODO: Confirm the overall date window from raw data: the dissertation scopes the study to 2010 to mid-2025 (p. 7), gives January 2018 to July 2025 for Twitter only (p. 21), states no restriction for Reddit (p. 22), and its forecast figures span 2010 to 2026 (pp. 38-39).] Collection was organised by an auditable keyword scheme with six seed-rule topics: taxation and fiscal policy; inflation, cost of living and prices; welfare and benefits; energy and utilities; housing and mortgages; and the labour market (dissertation p. 18; the scheme, including its disambiguation priority and negation window, is transcribed in Appendix C). Keyword tagging then assigned classified posts to nine analysis categories, of which an Other/Unclear category absorbed roughly a third (dissertation pp. 41-42). [TODO: Report exact per-category counts after the recount.] The recovered pipeline records settle the corpus total at the analysis level: the preprocessing log (5 August 2025) fetches 255,446 comments from the project database, and the executed notebook merges the same 255,446 rows,

Table 1: Dissertation-stage classifier comparison (dissertation Table 7.1, p. 35). The BERT and BiLSTM rows match the appendix confusion matrices exactly and therefore measure agreement with the VADER-seeded labels over the full classified corpus; the VADER and logistic regression rows appear to be scored against the 200-post human test set. The rows do not share a reference standard, and no row measures fine-tuned performance against the human criterion; the dissertation's own notebook output computes the BERT figures against the corpus labels (p. 31). [TODO: Recount: recompute every model against both reference standards, report them as separate columns, and attach binomial intervals to the n=200 criterion figures.]

Model	Accuracy	Macro precision	Macro recall	Macro F1
BERT (fine-tuned)	88.06%	0.8815	0.8744	0.8778
BiLSTM (GloVe)	81.51%	0.8141	0.8099	0.8111
VADER baseline	65.20%	0.66	0.65	0.65
Logistic regression (BoW)	71.40%	0.72	0.71	0.71

attached to 6,650 parent posts. The unit of analysis is therefore the comment, with parent posts available as a clustering level. [TODO: Report per-platform totals once the raw archive is located; the dissertation's per-platform figures (approximately 120k Twitter, 130k Reddit, 120k YouTube, 25k Quora; pp. 21-24) sum to roughly 395k, and the 279k headline (p. 25) matches neither those sums nor the 255,446 pipeline total, so both remain unexplained pending the archive.] Machine labels come from a fine-tuned BERT classifier (macro-F1 0.878 against the VADER-seeded corpus labels, not against the human criterion; dissertation Table 7.1, p. 35), trained on VADER-seeded labels; Table 1 reproduces the dissertation-stage comparison and Figure 1 the per-category sentiment shares it produced. We disclose this circularity and address it with the human criterion below and with the artefact control in Section 5. The dissertation-stage shares themselves fail face validity: every policy category is plurality positive for UK economic discourse spanning a cost-of-living crisis, which is substantively implausible and is best read as a candidate artefact of the lexicon-seeded labels. The expanded human criterion measures this directly; if it confirms a positive bias, that bias enters the generalizability and invariance analyses as a measured quantity rather than an assumption. The corpus is held fixed across all six steps, so every comparison uses the same posts.

Anchor events. Cross-platform comparison requires posts about the same event at the same time. We pre-identified a set of UK fiscal policy events observed on all four platforms, with the September 2022 mini-budget as the centrepiece, and defined two windows per event: an acute 48-hour window and an extended two-week window. [TODO: State the event-assignment rule: the keyword scheme of Appendix C plus the timestamp field per platform (question, answer, or comment time), fixed ex ante; validate relevance on a manual subsample within the expanded annotation.] We expect the acute windows to be sparse on Quora, whose question-and-answer format does not lend itself to event-locked measurement; the audit is designed to identify such platforms so they can be given a reduced role or excluded, and when a platform is excluded we report a gap in the data and claim nothing about its equivalence. An anchor-density audit precedes all analysis (Section 5); platforms that fail the minimum post-count threshold for an event-window are excluded from that comparison, and the analysis is restricted to the platforms with adequate density if necessary. The dissertation reports lower data density in early 2022 (p. 25), so density is checked per event-window rather than assumed.

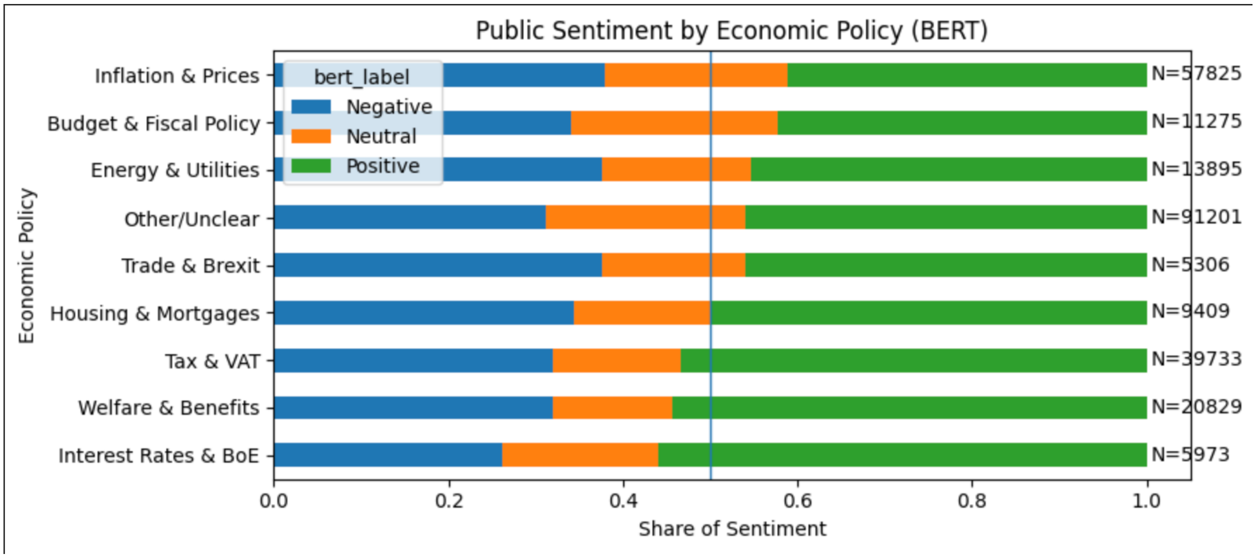

Figure 1: Sentiment shares by policy category under the fine-tuned BERT classifier, as they stood at dissertation stage (Figure 7.12, p. 41). The nine keyword-assigned categories carry per-category counts that sum to 255,446, one of the three corpus totals the recount must reconcile, and the Other/Unclear category absorbs roughly a third of classified posts.

Table 2: Anchor events with dates and per-platform post counts in the 48-hour and two-week windows.

[AWAITING the corpus recount and the anchor-density audit. Fix the inclusion rule ex ante: minimum posts per platform-event-window cell ([TODO: value, with justification]), applied to both windows; failing cells are excluded and listed here.]

Gold-standard sample. The dissertation validated machine labels on 200 manually annotated posts, balanced across the four platforms, with inter-annotator agreement of Cohen's $\kappa \approx 0.78$ (dissertation pp. 25, 28). Because that sample is small and was not drawn by probability sampling, we expand it to 1,000-1,500 posts drawn by stratified probability sampling, with strata defined by platform, anchor event, and predicted class, and inclusion probabilities retained for design-weighted estimates. A second annotator codes a [TODO: subset of n] posts; disagreements are adjudicated and inter-annotator reliability is reported. This sample is the criterion for the generalizability study and for validating the machine labels; the invariance models themselves are fitted on the machine-labelled corpus within anchor windows, with the criterion validating their inputs rather than forming their estimation sample. Because the criterion rests on human judgment rather than on a second lexicon, it breaks the circularity of the VADER-seeded training labels. Human validation remains essential for machine annotation [Pangakis and Wolken, 2025, Atreja et al., 2025].

Ethics. The underlying study was reviewed by the University of Portsmouth Faculty Ethics Committee (School of Computing) and classified as low risk (certificate TETHIC-2025-111094; dissertation pp. 15, 60), with legitimate interests as the lawful basis under UK GDPR (p. 14). We analyse public posts only, apply data minimisation, and process data under the Article 89 research safeguards. We report aggregates, paraphrase any quoted material, make no attempt to identify users, and share dehydrated identifiers rather than full text where platform terms require it. [TODO: Confirm identifier sharing is feasible: the dissertation committed to deleting raw data after the project and retaining only anonymised aggregates (p. 15), user identifiers were hashed (p. 25), and the Quora licence prohibits full-text reproduction (p. 24).]

4 Step 1: Establishing Divergence

Before asking whether the platform shapes the measurement, we establish that there is something to explain. For each anchor event and window, we compute the distribution of machine-assigned sentiment labels per platform and compare platforms pairwise. Because the comparison is confined to a shared event window, it constrains topic and time far more tightly than whole-corpus comparison, though different sub-events may still dominate on different platforms within a window; within-window composition is handled in Step 4. We test differences in class proportions with chi-square tests and report bootstrap confidence intervals, clustered by user or thread where hashed identifiers permit, and Cramér's V, since at this sample size significance alone is uninformative and interpretation rests on the effect sizes. Divergence at this step is only a precondition. It is consistent both with genuinely different populations and with platform-dependent measurement, and the remainder of the paper separates the two. Prior work stops at this step or assumes it away [Pellert et al., 2022, Overgaard et al., 2024].

[AWAITING Results: per-event divergence estimates and effect sizes, pending the corpus recount and anchor-density audit.]

5 Steps 2-4: Reliability, Invariance, Adjustment

We treat the classifier as a measurement instrument and the platform as a candidate component of the measurement procedure [Jacobs and Wallach, 2021, Halterman and Keith, 2026]. The design proceeds in six steps: establish divergence (Section 4); estimate reliability per platform with a generalizability study; test measurement invariance and differential item functioning with platform as the grouping facet; adjust for composition and timing in a multilevel model; link scales where the evidence permits (Section 6); and quantify the cost of skipping the first five steps (Section 7). As far as we are aware, this is the first application of the psychometric equivalence toolkit [Cronbach et al., 1972, Meredith, 1993, Brennan, 2001, Kolen and Brennan, 2014] to machine-classified policy sentiment with platform as the grouping facet.

Two design commitments apply to every step that follows. The anchor-density audit comes first: every event-window-platform cell must meet a minimum post count before it enters any model, and if cells fail, the study narrows to the dense subset of platforms rather than papering over sparsity [Morstatter et al., 2013]. And platform is a fixed facet throughout: our claims concern these four platforms, not a universe of platforms from which they were sampled.

5.1 Step 2: Fixed-facet generalizability study

The criterion is the gold-standard sample of Section 3. For each platform m we estimate a generalizability study with posts nested in events and coders as a random facet, analysed on pre-adjudication ratings; the executed design is unbalanced, since the second coder rates a subset, and components are estimated by REML with interval estimates rather than point comparisons. The classifier enters the same decomposition as an additional fixed rater, so machine-human disagreement is an estimable per-platform component:

$$X_{p(e)c}^{(m)} = \mu^{(m)} + \nu_e^{(m)} + \nu_{p(e)}^{(m)} + \nu_c^{(m)} + \nu_{ec}^{(m)} + \nu_{p(e)c,\varepsilon}^{(m)}, \tag{1}$$

with variance components for events, posts within events, coders, their interactions, and residual estimated separately at each fixed level m [Cronbach et al., 1972, Brennan, 2001]. Comparing absolute error components across platforms shows whether the measurement procedure is equally dependable on every platform, something a single reliability coefficient cannot show. Dependability coefficients are population-bound, so cross-platform coefficient comparisons are reported as descriptive only. Generalizability designs have recently modelled prompts and models as facets of automated annotation [Camuffo et al., 2026], and annotator-facet studies exist [Kennedy et al., 2020, Sachdeva et al., 2022], but none of them models the platform itself.

5.2 Step 3: Invariance and DIF with platform as the group

We fit multiple-group measurement models grouped by platform. The indicators are binary presence codes for each of the eight NRC emotions (Figure 2), derived from the raw lexicon match counts rather than a single dominant label (the executed notebook already records these counts per comment; Figure 10), together with the three-category sentiment class treated as ordinal under the threshold parameterisation. Dimensionality is tested per platform before any invariance constraint is imposed, and the invariance battery is conditional on an acceptable configural model. Because the NRC indicators derive from a lexicon, they are excluded from the non-lexicon replication arm, which is restricted to the sentiment indicator set. Four safeguards apply at this step. Identification follows the threshold parameterisation for categorical indicators [Wu and Estabrook, 2016]; we do not treat categorical labels as linear indicators. Invariance decisions rest on effect-size criteria, changes in CFI and RMSEA [Chen, 2007, Putnick and Bornstein, 2016]; chi-square values are reported but decisions do not rest on them, since at this sample size the test rejects trivial differences. Because these cutoffs were calibrated for continuous indicators under maximum likelihood, we treat them as heuristics subject to sensitivity analysis under the categorical estimator, and per-event DIF results aggregate into a platform-level claim by a rule fixed before estimation [TODO: state the rule: an indicator-platform pair is flagged only when non-invariance appears in at least k of the adequately powered events in both classifier arms; choose k and the power criterion]. DIF is tested separately within each anchor event, so an event-specific artefact cannot be mistaken for a global platform effect. And because the choice of anchors can itself build in the conclusion, we re-estimate DIF anchor-free [Chen et al., 2023] and run alignment optimisation [Muthén and Asparouhov, 2018] alongside the constrained models. These are three identification strategies for the same model rather than independent tests, so agreement rules out anchor-choice artefacts and nothing more; a claim of non-invariance must survive all three, a claim of invariance must do the same, and disagreement is reported as inconclusive.

5.3 Step 4: Multilevel adjustment

Steps 2 and 3 could still confound measurement with composition. We therefore fit a multilevel model with posts nested in events and platform as a fixed effect. Time since event enters the model explicitly: platforms differ systematically in when their users respond, and this timing component may reflect real opinion dynamics as well as affordance. We therefore estimate platform contrasts with and without the latency term and report the pair, as total and direct platform differences, alongside rather than inside the measurement comparison. Where the multilevel structure permits,

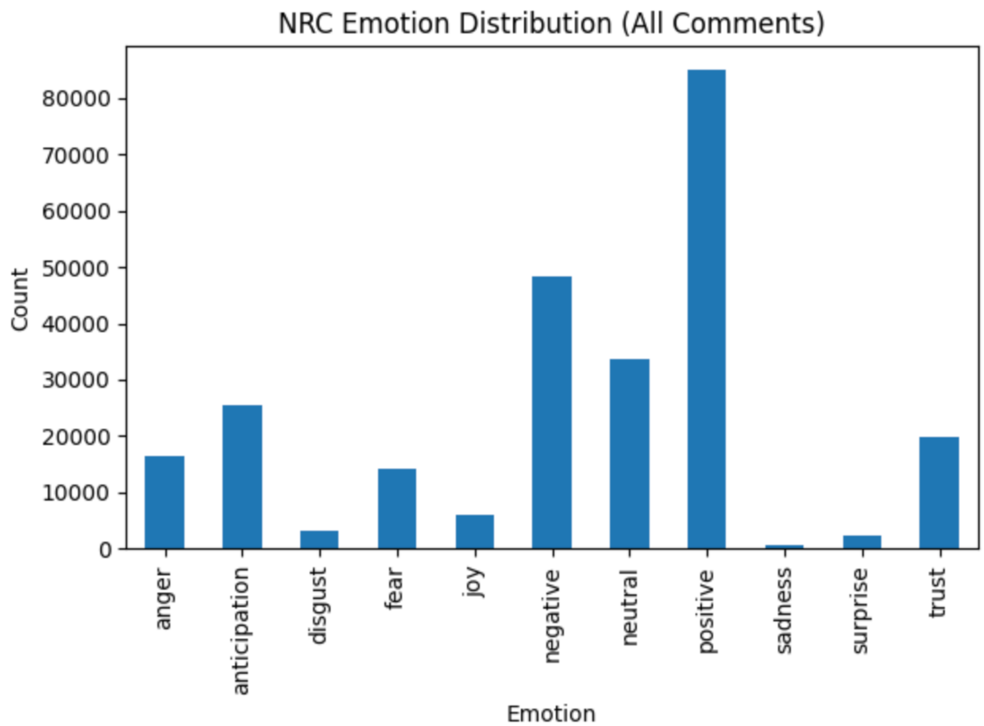

Figure 2: Dissertation-stage NRC distribution over all comments (dissertation Figure 7.8, p. 40): eight emotion categories plus positive, negative, and neutral tallies. The bars sum to the classified corpus, so each comment carried one mutually exclusive label at dissertation stage; the recomputation replaces this single-label assignment with binary presence codes per emotion, which become the Step 3 indicators.

we also apply the cluster-bias test of Jak et al. [2013], which examines invariance violations across clusters directly.

5.4 Classifier-artefact control and interpretive stance

A platform effect estimated with one classifier could be an artefact of that classifier, and our training labels descend from a lexicon, as already disclosed. Every analysis is therefore replicated with a non-lexicon transformer [TODO: name the instrument and its label provenance; it must have no VADER ancestry, for example an off-the-shelf transformer fine-tuned on independently human-labelled sentiment data], and only convergent non-invariance, the same pattern under both classifier families, is interpreted as platform-level non-equivalence rather than a lexicon artefact [Czarnek and Stillwell, 2022]. Convergence cannot rule out biases shared by both instrument families, a residual limitation stated in Section 8. Divergent patterns are reported as classifier artefacts, which also constrains the analyst degrees of freedom that make automated-annotation pipelines fragile [Baumann et al., 2025, Atreja et al., 2025]. The two families do behave differently: at dissertation stage the two families agreed on 88.1 per cent of comments (225,010 of 255,446; executed notebook output, and Figure 9), a figure the dissertation text misreports as 85 per cent, with disagreement varying modestly by topic, from roughly 10 to 13 per cent and highest for Trade and Brexit (Figures 3 and 4). The convergence criterion relies on this variation. The consequence for measurement is visible in Figure 5: rescoring the same corpus with the BiLSTM preserves the

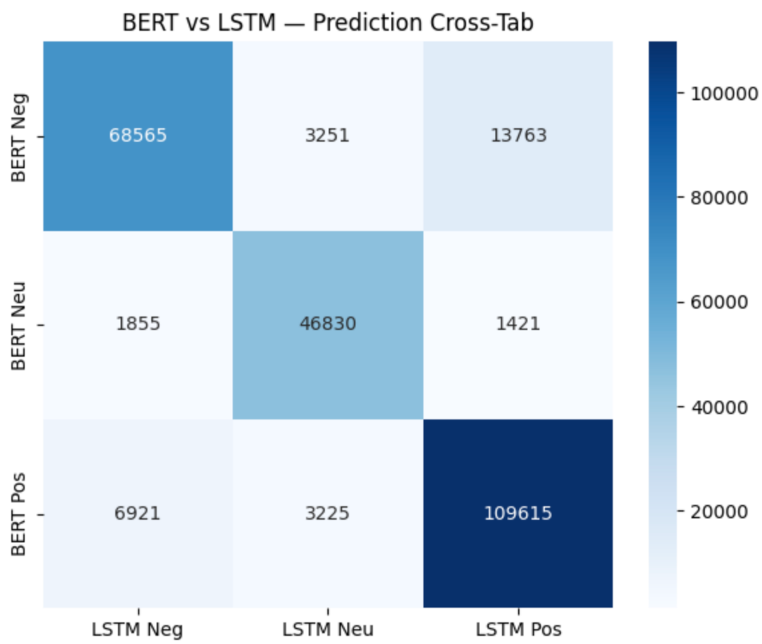

Figure 3: Dissertation-stage cross-tabulation of BERT and BiLSTM predictions over the classified corpus (dissertation Figure 7.15, p. 43). Both families were trained on the same VADER-seeded labels, so their disagreement illustrates instrument-dependence within a shared label heritage; the BiLSTM is not the non-lexicon control, which is specified in the text.

ranking of policy categories but changes the measured sentiment shares, a within-platform instance of the instrument-dependence this paper tests between platforms.

Finally, invariance evidence is diagnostic, not a licence [Robitzsch and Lüdtke, 2023]. Passing a threshold does not certify pooled scores as interchangeable; it localises where comparability holds and where it fails, and every claim in Sections 6 and 7 is conditioned on that localisation. The framework thereby formalises the platform-affordance error that Sen et al. [2021] describe but do not test, and reaches a failure mode that label-error corrections do not touch, because those correct classifier error against a criterion within a platform, not non-equivalence between platforms [Egami et al., 2023].

6 Step 5: Linking

If partial invariance holds, we link the platform scales. What we attempt is linking rather than equating: equating requires that transformed scores be interchangeable, a standard machine-labelled sentiment across platforms cannot meet, so we settle for the weaker relation [Kolen and Brennan, 2014]. The linked object is a platform-level quantity, the platform-specific latent means and variances under the partial-invariance model, not any per-post score. The pre-identified fiscal events act as common stimuli rather than common items: different posts by different authors are scored on each platform, which in Kolen and Brennan's taxonomy is a moderation design, the weakest linking category, and we label it as such. The link is identified only under the assumption that latent sentiment toward each anchor event is equal in expectation across platform populations after the Step 4 composition adjustment; where that assumption is doubted, the crosswalk removes population differences along with measurement differences, and we report both indices for that

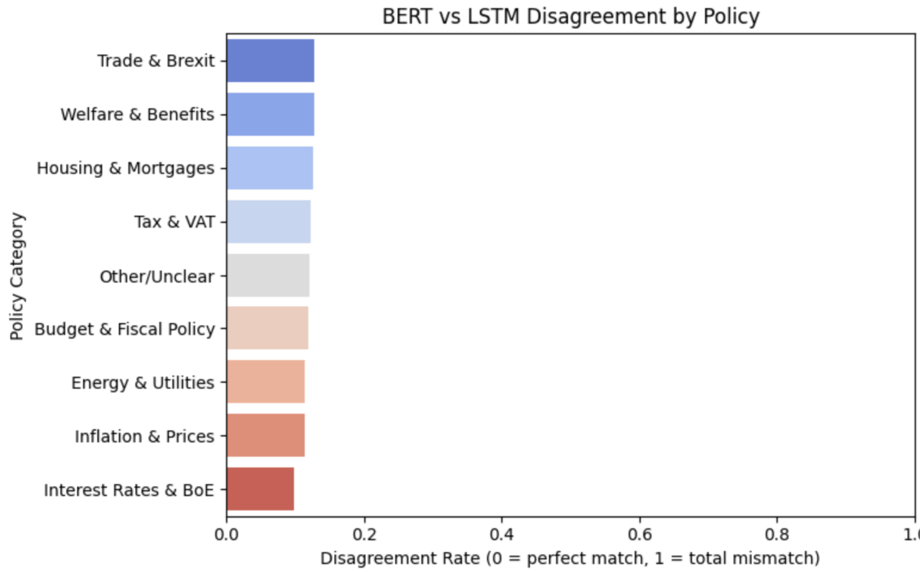

Figure 4: BERT-versus-BiLSTM disagreement rate by policy category at dissertation stage (Figure 7.14, p. 43). Disagreement is highest for Trade and Brexit and Welfare and Benefits, though the spread across categories is modest, roughly 10 to 13 per cent.

reason. The September 2022 mini-budget is the principal anchor, with the 48-hour and two-week windows defining the linking samples. Purification is ex ante: the rule for screening anchors for differential functioning is fixed before any linking function is estimated [TODO: state the purification statistic and threshold here; an unstated rule cannot be ex ante], so anchors cannot be chosen in a way that favours a particular result. For sensitivity we re-estimate the linking function dropping each anchor event in turn and report the spread; a link that depends on a single event is reported as such; we do not smooth over it.

The protocol does not presuppose that linking will succeed. If Step 3 shows non-invariance too severe to support even partial-invariance linking, we do not link. The deliverable then becomes platform-specific sentiment series plus the demonstration that a pooled cross-platform index is unsupported for this construct. We would report either outcome: a crosswalk with quantified uncertainty, or evidence that none exists for this construct.

[AWAITING Results: linking functions, purification outcomes, and leave-one-event-out sensitivity, pending Steps 1-3.]

7 Step 6: The Cost of Assuming Equivalence

The final step quantifies the cost of assuming equivalence, an assumption the field currently makes without testing it. From the same corpus we construct two indices. The naive index pools posts across platforms, the construction canonical since O'Connor et al. [2010]; the dissertation's version was a daily mean of scored posts (Figure 6), and Step 6 constructs both that variant and the share-negative variant. The adjusted index applies the platform-specific measurement parameters from Step 3 and, where the fork in Section 6 permits, the linking functions from Step 5. We compare the two on the quantities an analyst would actually report: index levels, trends, and the estimated sentiment shift around each anchor event, including sign flips in that shift. Both indices are also fed to a Prophet forecasting model, mirroring the dissertation's forecasting stage, which

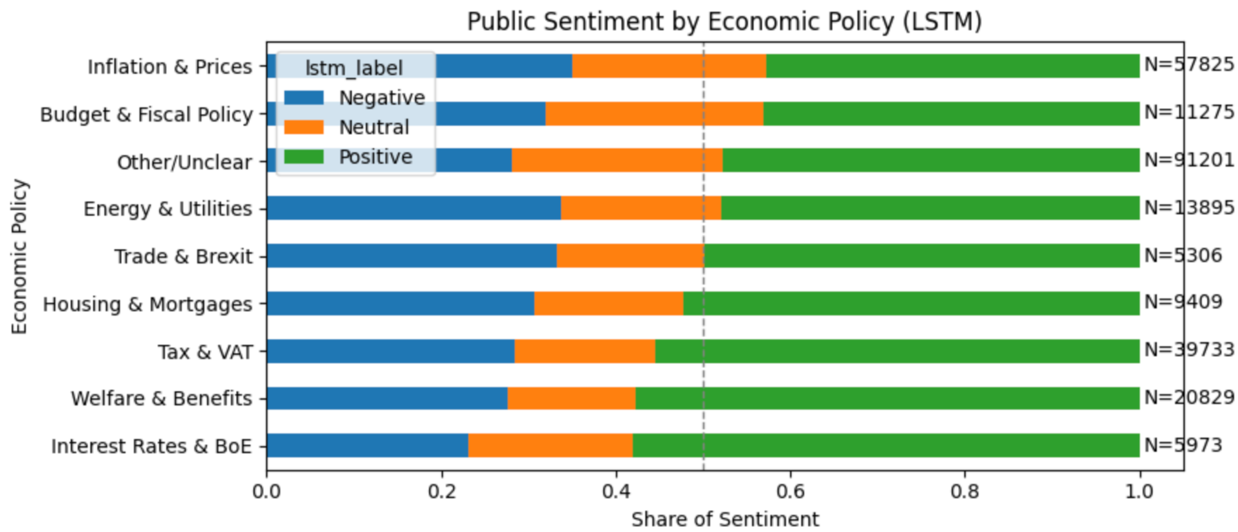

Figure 5: The same corpus rescored with the BiLSTM (dissertation Figure 7.13, p. 42). The broad category ordering is nearly preserved, with adjacent categories swapping rank; measured shares are not preserved. Compare Figure 1.

applied Prophet to a smoothed daily sentiment index and decomposed it into trend and seasonal components (Figure 7); the forecast comparison is a downstream-consequences illustration only, since a biased but stable index can forecast well, and the primary loss is index levels, event shifts, and sign flips. Because Steps 2, 3 and 5 estimate parameters within anchor windows, applying them to the full corpus is an extrapolation; we report a sensitivity analysis excluding the Other/Unclear category, which never enters an estimation window. [TODO: Specify how the Step 3 parameters become an index: the current plan is platform-specific latent means under the retained partial-invariance constraints, aggregated with fixed platform weights; confirm.] Finally, we simulate composition change: reweighting platform shares to mimic a platform losing API access [Davidson et al., 2023, Goanta et al., 2026] shows how much apparent sentiment movement is a measurement and composition artefact rather than opinion change. Corrections for label error [Egami et al., 2023] leave this quantity unchanged: they address disagreement between classifier and criterion, whereas the artefact here arises between platforms.

[AWAITING Results: naive versus adjusted index comparison, event-shift bias, sign flips, forecast error, and composition simulation.]

8 Limitations

Our claims are subject to five limitations. First, the selection confound. Users choose their platforms, and the same-event design controls topic and time, not who is posting. Any non-invariance we detect therefore bundles population composition with measurement mode; because the two can offset as well as compound, the bundle cannot be read as a bound on the pure mode effect, only as the platform-level difference in how the construct is measured [Sen et al., 2021]. The convergence criterion [Czarnek and Stillwell, 2022] removes classifier artefacts from this bundle; it cannot remove selection. Separating the two would require within-user cross-posting data, which we do not have.

Second, there is the question of platform scope. We study four platforms, and the anchor-density

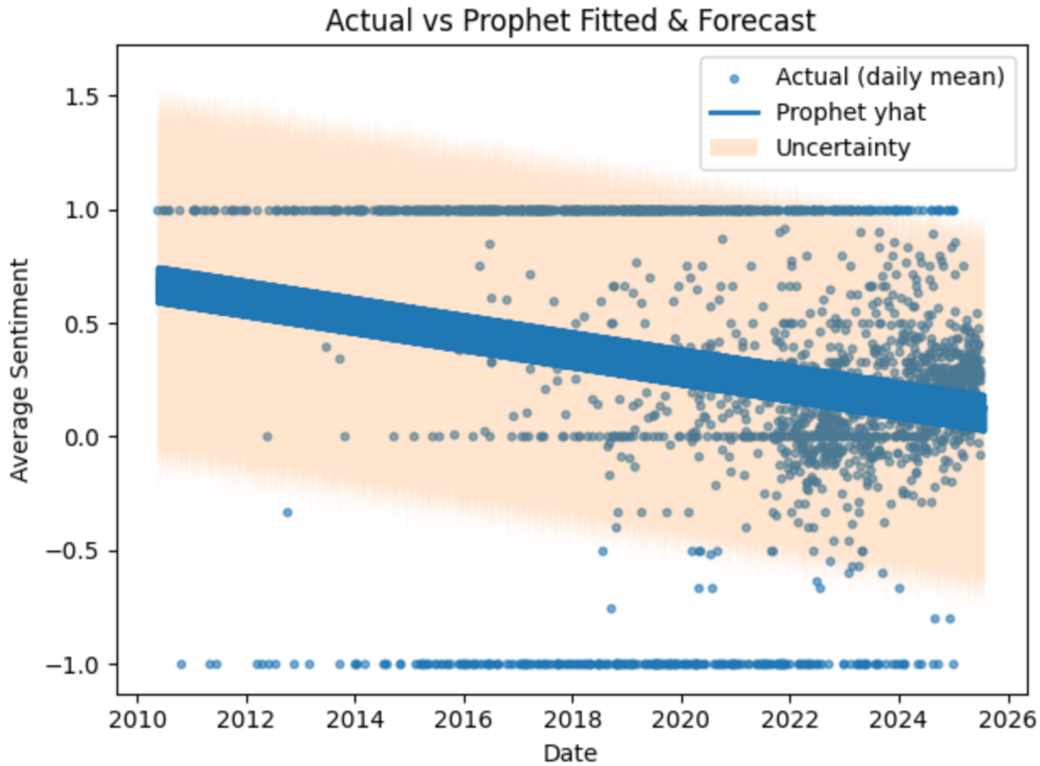

Figure 6: The naive pooled index in practice: dissertation-stage daily mean sentiment pooled across platforms, with the Prophet fit and forecast band (dissertation Figure 7.7, p. 39). Several problems with the construction are visible here: days with a single post pin the daily mean at plus or minus one, no support threshold or volume weighting is applied, the uncertainty band ignores the bounded scale, and the axis spans 2010 to 2026, one reason the collection window is under recount. Step 6 compares this construction with a measurement-adjusted alternative.

audit may narrow the analysis to the subset with sufficient same-event coverage. All findings are conditional on the retained platforms and events. Exclusion of a sparse platform is a data limitation, not evidence that it measures equivalently. Likewise, following Robitzsch and Lüdtke [2023], we read invariance results as diagnostic evidence about this corpus, not as a general warrant to pool.

Third, the construct itself. Our instrument measures expressed sentiment, not policy support. Negative sentiment in a thread about inflation need not be opposition to any policy [O'Connor et al., 2010]. Our equivalence results concern the sentiment measure itself; any mapping to attitudes would need separate validation.

Fourth, a single country and language. All data are UK-focused and English. Machine scoring is language-dependent [Nogara et al., 2025], and platform affordance mixes differ across national ecologies, so we make no cross-country claim.

Finally, provenance. The dissertation underlying this paper reported several internally inconsistent statistics: per-platform counts summing to roughly 395,000 against a 279,000 headline (pp. 21-25), a forecast MAE of approximately 0.02 in the text but 0.05 in the goal table (pp. 39, 46-47), post-calibration ECE of 0.05 in the methodology but 0.03 in the results while the reliability diagrams themselves embed ECE values of 0.534 for BERT and 0.012 for BiLSTM, contradicting both (pp. 30, 36-37), manual labelling coverage of 0.07% in one chapter and approximately 0.5% in

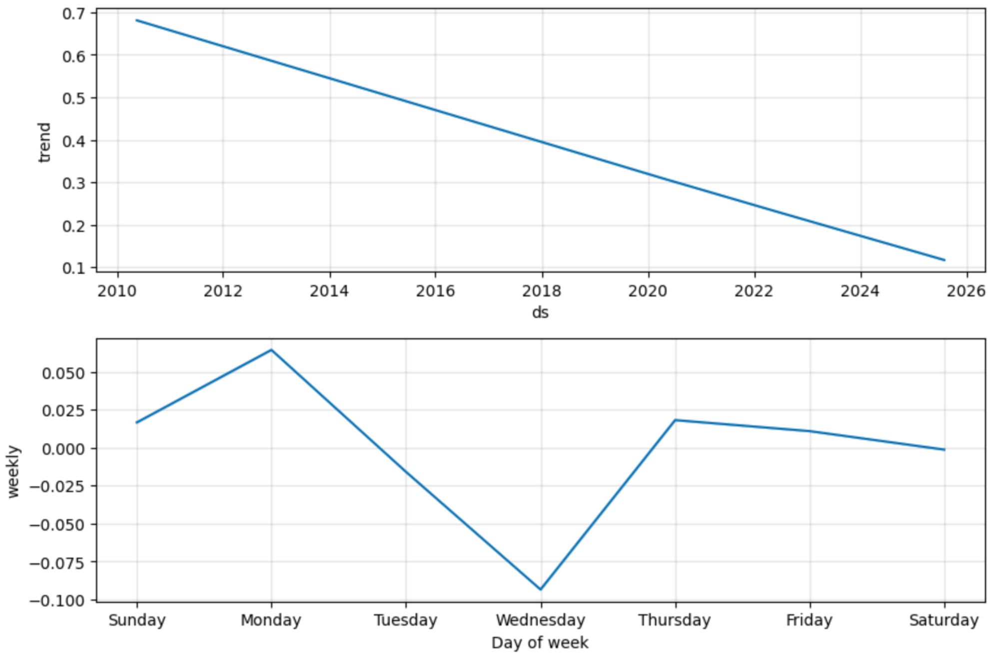

Figure 7: Dissertation-stage Prophet decomposition of the pooled index (dissertation Figure 7.6, p. 39): a declining long-run trend and a midweek dip that Step 6 treats as a candidate composition artefact, equally consistent with weekday differences in platform mix and with mood itself. If Step 3 finds non-invariance, components like these inherit the measurement artefacts of the pooled construction.

another (pp. 25, 45), an accuracy figure appearing as both 88.06% and approximately 0.89 (pp. 35, 50), and a forecast figure titled with a platform the study excluded (p. 38). The recovered execution records add two more: the dissertation text reports 85 per cent model agreement where the executed notebook computes 88.1 per cent, and the reproducibility appendix describes a temporal train-test split and older library versions where the executed code applies a stratified random 90/10 split with seed 42 under newer libraries. The dissertation-stage figures reproduced in this draft are included as evidence of the instrument as it stood; every figure will be regenerated in the recomputation, and the inconsistent reliability diagrams are deliberately not reproduced. [TODO: List corrected totals and recomputed values.] All statistics in the final version will be recomputed from the retained archive before submission, and the recomputed figures will supersede the dissertation throughout. [TODO: State the retention basis for the raw archive. The analysis code, pipeline logs, and fully executed notebook were recovered on 21 August 2026; the project database (a local MySQL instance at run time) and the intermediate JSON extracts are not on the analysis machine and must be located, or the corpus re-collected, before any recomputation. The dissertation committed to deleting raw data after project completion (pp. 15, 25), which this must reconcile.]

9 Ethics Statement

This is an observational study of public posts on public policy topics. Collection was approved under the University of Portsmouth ethics review process (School of Computing, Faculty Ethics Committee route; certificate TETHIC-2025-111094, application 27 May 2025, supervisor sign-off 1 June 2025), which classified the project as low risk and complies with GDPR: no private, deleted, or direct-message content was collected, released artefacts contain no personal identifiers, and quoted examples are paraphrased. The dissertation-stage criterion set comprised 200 manually labelled posts balanced across platforms, with inter-annotator agreement of approximately 0.78 (Cohen's kappa; dissertation p. 28). Annotators were [TODO: recruitment and compensation details]. [TODO: Confirm whether the expanded annotation round and continued data retention are covered by an amendment to TETHIC-2025-111094 or a new approval, and cite the reference number.] The main foreseeable misuse is reading our diagnostics as a ranking of platforms by accuracy; they test equivalence and do not score quality, and we caution against that reading.

10 Conclusion

Until shown otherwise, the platform should be treated as part of the measurement instrument. We formalise that claim and test it with the psychometric equivalence toolkit, applied, we believe for the first time, to machine-labelled policy sentiment with the platforms as the groups. Whatever the result, something useful follows: if invariance holds we will know where pooling can be defended, and if it fails we will have an estimate of where, and by how much, pooling misleads. [TODO: One-sentence headline finding once Step 3 and Step 6 results are in.]

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A Robustness Battery

Every headline result is paired with the checks below. [AWAITING Full battery tables pending Round 3 recomputation.]

Invariance and DIF. Categorical-indicator models with threshold constraints, judged on effect-size criteria rather than chi-square. Alongside the constrained models we run alignment [Muthén and Asparouhov, 2018] and anchor-free DIF [Chen et al., 2023], so no conclusion depends on a hand-picked anchor set.

Classifier-artefact control. The full test suite is replicated with a non-lexicon transformer. A result counts as a platform mode effect only when both classifier families show it [Czarnek and Stillwell, 2022]; a pattern present under one family alone is recorded as a classifier artefact.

Linking sensitivity. Leave-one-event-out re-estimation of linking constants over the pre-identified fiscal-event anchors, with the September 2022 mini-budget as the main anchor; both 48-hour and 2-week event windows; anchor purification fixed and applied before any estimation.

Uncertainty and calibration. Bootstrap confidence intervals on all indices; McNemar tests for classifier comparisons; temperature-scaled calibration checked per platform. At dissertation stage the executed notebook records McNemar $b = 22{,}429$, $c = 5{,}680$, $p < 0.001$, and a bootstrap macro-F1 difference of 0.067 (95 per cent interval 0.065 to 0.068) between the two classifier families, against the VADER-seeded reference labels.

Annotation reliability. Expansion of the human criterion sample from the dissertation's 200 balanced posts (Cohen's kappa approximately 0.78; dissertation p. 28) to [TODO: 1,000-1,500] probability-sampled posts with a second-annotator subset; agreement [TODO: kappa].

B Dissertation-Stage Diagnostic Outputs

The dissertation-stage per-class diagnostics are reproduced here in their original form (Figures 8 and 11), recording the state of the instrument at dissertation stage; they are console outputs rather than typeset figures and will be regenerated, with the corrected reliability diagrams, in the recomputation. [TODO: Regeneration checklist: transcribe both console reports into one typeset table noting the VADER-seeded reference labels; identical category order across paired figures; rename LSTM to BiLSTM; legends in plain English, placed outside the axes; sequential palette for unsigned rates; axes limited to the data range; state the Prophet interval level; vector output, no screenshot borders. The checklist applies to the dissertation-stage figures in the main text as well.] Both classifiers struggled most with the neutral class; the expanded criterion sample is designed to measure this pattern properly. Figure 9 shows the executed notebook's agreement tally between the two families, the source of the 88.1 per cent figure in Section 5, and Figure 10 an example of the per-comment NRC match counts from which the Step 3 binary indicators derive.

```
📊 Classification Report (per class):
              precision    recall  f1-score   support

    negative       0.86      0.86      0.86     85305
     neutral       0.90      0.86      0.88     52383
    positive       0.89      0.91      0.90    117758

    accuracy                           0.88    255446
   macro avg       0.88      0.87      0.88    255446
weighted avg       0.88      0.88      0.88    255446


🔢 Confusion Matrix:
          pred_neg  pred_neu  pred_pos
negative     73486      2187      9632
neutral       3980     44873      3530
positive      8113      3046    106599
```

Figure 8: Dissertation-stage per-class report and confusion matrix, fine-tuned BERT (dissertation Figure 7.1, p. 36). Class supports total 255,446; the reference labels are the VADER-seeded corpus labels, so these diagnostics measure agreement with the seed lexicon, not accuracy against the human criterion.

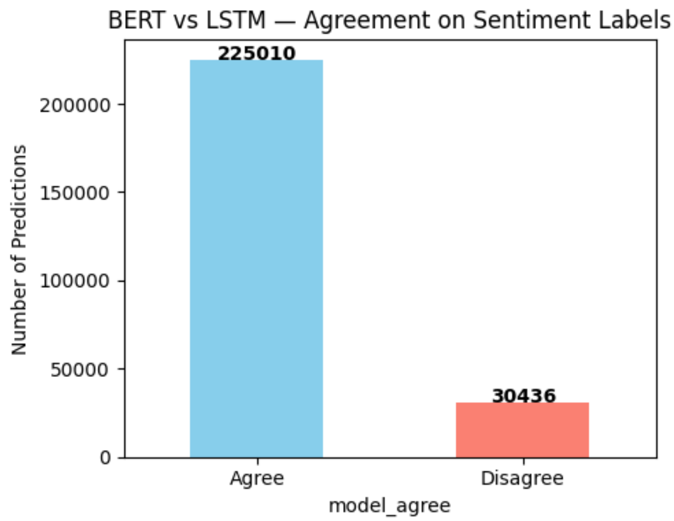

Figure 9: Agreement between the two classifier families over the classified corpus, from the executed notebook: 225,010 of 255,446 comments (88.1 per cent). The dissertation text's figure of 85 per cent is superseded.

C Collection Keyword Scheme

The dissertation's auditable seed rules (p. 18), reproduced here in typeset form, subject to the recount. Six topics, each with a keyword list and, for the first two, regular-expression hints: taxation and fiscal policy (tax, VAT, income tax, NI, national insurance, HMRC, budget, chancellor, Finance Bill, fiscal, levy); inflation, cost of living and prices (inflation, CPI, prices, cost of living, food prices, fuel prices, energy bills, price cap, BoE, interest rate); welfare, benefits and Universal Credit (welfare, benefits, Universal Credit, UC, DWP, benefit cap, means-tested, child benefit); energy and utilities (energy price guarantee, Ofgem, electricity bill, gas bill, standing charge, fuel duty); housing and mortgages (mortgage, rent, landlord, housing benefit, stamp duty, help to buy); and the labour market (minimum wage, national living wage, pay rise, pay freeze, strike, unemployment). When more than one topic matched, priority ran taxation, then inflation, welfare, energy, housing, labour. Negation terms within three tokens of a polarity keyword flagged the post for review. A gazetteer of institutions (HM Treasury, Bank of England, DWP, Ofgem, OBR) and role mappings (Chancellor, Prime Minister, Shadow Chancellor) supplied entity hints.

D Data and Code Availability

Code, configuration, annotation guidelines, label sets, and aggregated per-event statistics are available at [TODO: repository URL]. The dissertation stage ran on Google Colab (Python 3.11); the executed notebook's environment (torch 2.8, transformers 4.55, spacy 3.8, nltk 3.9, pandas 2.2) differs from the versions listed in the dissertation's reproducibility appendix (p. 19), and the recomputation will pin the real environment. The executed split was stratified random, 90 per cent train and 10 per cent test (229,901 and 25,545 comments, seed 42), not the temporal split the reproducibility appendix describes; the recomputation will use and report a temporal split alongside it. Raw post text is not redistributed, in line with platform terms; post identifiers are released where terms permit rehydration. Released artefacts contain no personal data. Compute used: a

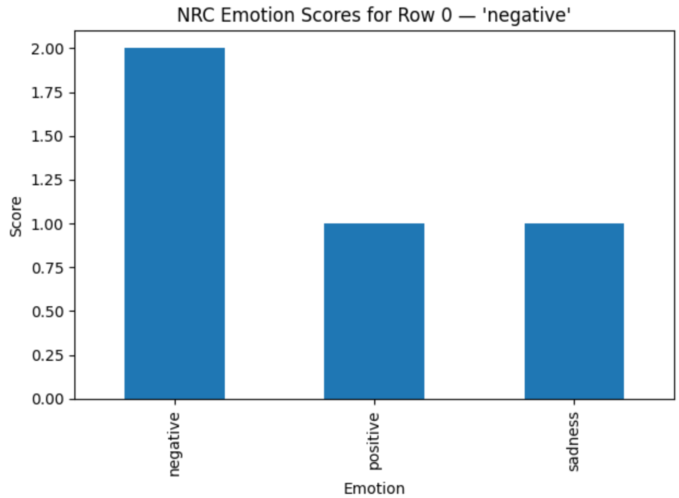

Figure 10: Per-comment NRC lexicon match counts for one example (executed notebook). These raw counts, not the single dominant label, are the basis of the Step 3 binary emotion indicators.

single NVIDIA T4 GPU (16 GB) on Google Colab Pro for the dissertation-stage models (dissertation p. 19); [TODO: GPU hours, and hardware for the recomputation]. For the conference version, the completed mandatory ICWSM checklist will be placed after the references; together with the above, this covers the reproducibility items for ML experiments and data assets.

📊 Classification report (full data):

	precision	recall	f1-score	support
negative	0.81	0.74	0.77	85305
neutral	0.81	0.83	0.82	52383
positive	0.82	0.87	0.84	117758
accuracy			0.82	255446
macro avg	0.81	0.81	0.81	255446
weighted avg	0.81	0.82	0.81	255446

🔢 Confusion matrix (full data):

	pred_neg	pred_neu	pred_pos
negative	62837	4449	18019
neutral	4435	43270	4678
positive	10069	5587	102102

Figure 11: Per-class report and confusion matrix for the BiLSTM at dissertation stage (Figure 7.2, p. 36). The reference labels are again the VADER-seeded corpus labels, not the human criterion.